1 $e^{ix} = \cos x + i \sin x.$

$$e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} = 1 + ix + \frac{(ix)^2}{2} + \frac{(ix)^3}{6} + \frac{(ix)^4}{4!} + \dots =$$

= (separate into $n=$ <u>even</u> and $n=$ <u>odd</u> series)

$$= \sum_{\substack{n=2m \\ m=0}}^{\infty} \frac{(ix)^{2m}}{(2m)!} + \sum_{\substack{n=2k+1 \\ k=0}} \frac{(ix)^{2k+1}}{(2k+1)!}$$

$$(ix)^{2m} = i^{2m} x^{2m} = (i^2)^m x^{2m} = (-1)^m x^{2m}$$

$$(ix)^{2k+1} = i^{2k+1} x^{2k+1} = i (i^2)^k x^{2k+1}$$
$$= i (-1)^k x^{2k+1}$$

$$\Rightarrow e^{ix} = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!} + i \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!} =$$

$$= 1 - \frac{x^2}{2} + \frac{x^4}{4!} + \dots + i\left(x - \frac{x^3}{3!} + \dots \right) =$$

$$= \cos x + i (\sin x).$$

(2) Solve $\cos(z) = 2$

$$\frac{1}{2}\left(e^{iz} + e^{-iz}\right) = 2$$

$$e^{iz} + e^{-iz} = 4$$

$$e^{i2z} + 1 = 4e^{iz}$$

$$\Rightarrow \left(e^{iz}\right)^2 - 4e^{iz} + 1 = 0$$

solving quadratic equation:

$$e^{iz} = 2 \pm \sqrt{2^2 - 1} = 2 \pm \sqrt{3}$$

$$\Rightarrow iz = \ln(2 \pm \sqrt{3}) + 2\pi N i$$

any integer $\left(e^{i2\pi N} = 1\right)$

$$z = -i \ln(2 \pm \sqrt{3}) + 2\pi N$$

Thus $z_1 = -i \ln(2+\sqrt{3}) + 2\pi N$

$$z_2 = -i \ln(2-\sqrt{3}) + 2\pi N$$

z_2 can be re-written because

$$\ln(2-\sqrt{3}) = \ln \frac{(2-\sqrt{3})(2+\sqrt{3})}{2+\sqrt{3}} = \ln \frac{4-3}{2+\sqrt{3}} = -\ln(2+\sqrt{3})$$

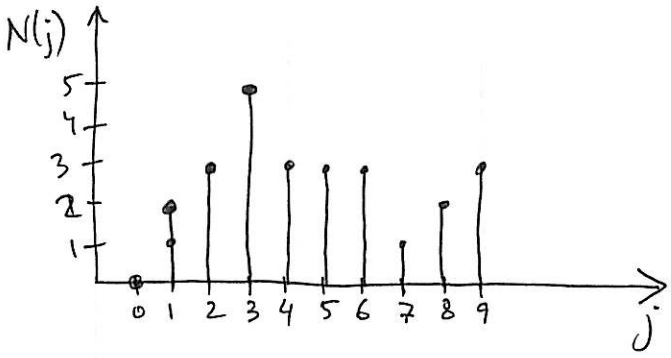
$$\Rightarrow z_2 = +i \ln(2+\sqrt{3}) + 2\pi N$$

Thus $z_{1,2} = \pm i \ln(2+\sqrt{3}) + 2\pi N$.

P1.10 $\pi = 3.14159265358979323846264 3\ldots$

(you can find many more digits from, for example, wikipedia.org if you don't have Maple or Mathematica programs)

(a) $N = 25$ digits total, of which 0 is encountered 0 times



1	" — "	2
2	" — "	3
3	" — "	5
4	" — "	3
5	" — "	3
6	" — "	3
7	" — "	1
8	" — "	2
9	" — "	3

Thus probability of getting them are: $\left(P(j) = \frac{N(j)}{N}\right)$.

$P(0) = 0 \quad P(1) = 2/25 \quad P(2) = 3/25 \quad P(3) = 5/25 = 1/5$

$P(4) = 3/25 \quad P(5) = 3/25 \quad P(6) = 3/25 \quad P(7) = 1/25 \quad P(8) = 2/25 \quad P(9) = 3/25$

(b) Most probable is number 3 : $P(3) = \frac{1}{5}$

Median is number 4 : out of 25 total, 13 outcomes result in number ≤ 4 and 12 result in ≥ 5.

Average : $\langle j \rangle = \sum_{j=0}^{9} j\, P(j) = \frac{1}{25}\Big(0\cdot 0 + 1\cdot 2 + 2\cdot 3 + 3\cdot 5 +$

$+ 4\cdot 3 + 5\cdot 3 + 6\cdot 3 + 7\cdot 1 + 8\cdot 2 + 9\cdot 3\Big) = \frac{118}{25} = 4.72$

(c) $\langle j^2 \rangle = \sum_{j=0}^{9} j^2 P(j) = \frac{1}{25}\Big(0\cdot 0 + 1^2\cdot 2 + 4\cdot 3 + 9\cdot 5 + 16\cdot 3 +$

$+ 25\cdot 3 + 36\cdot 3 + 49\cdot 1 + 64\cdot 2 + 81\cdot 3\Big) = \frac{710}{25} = 28.4$

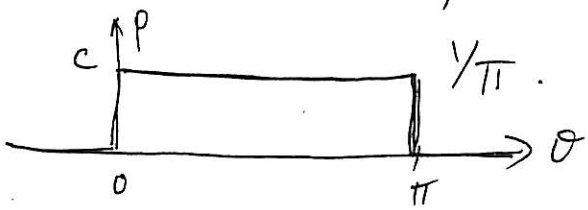
$\sigma^2 = \langle j^2 \rangle - \langle j \rangle^2 = 28.4 - (4.72)^2 = 6.12$

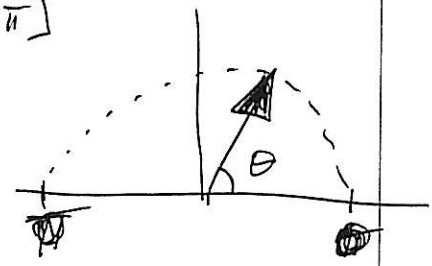
$\Rightarrow \sigma = \sqrt{6.12} = 2.47$

P1.11 Speedometer needle is free to swing and is equally likely to come to rest at any angle θ between 0 and π.

(a) Find $p(\theta)$.

$$p(\theta) = \begin{cases} \text{const} = C, & \text{for } \theta \in [0, \pi] \\ 0, & \text{otherwise} \end{cases}$$





We find C via: $\int_0^\pi p(\theta)\, d\theta = 1 \Rightarrow C \cdot \pi = 1$

$C = 1/\pi$

(b) $\langle \theta \rangle = \int_0^\pi \theta \frac{1}{\pi} d\theta = \frac{1}{\pi} \frac{\theta^2}{2}\Big|_0^\pi = \frac{\pi^2}{2\pi} = \frac{\pi}{2}$ (naturally)

$\langle \theta^2 \rangle = \frac{1}{\pi} \int \theta^2 d\theta = \frac{1}{\pi} \frac{\theta^3}{3}\Big|_0^\pi = \frac{\pi^2}{3}$

$\sigma^2 = \frac{\pi^2}{3} - \left(\frac{\pi}{2}\right)^2 = \pi^2 \left(\frac{1}{3} - \frac{1}{4}\right) = \pi^2 \frac{4-3}{12} = \frac{\pi^2}{12}$

$\Rightarrow \sigma = \pi/\sqrt{12} = \frac{\pi}{2\sqrt{3}}$.

(c) $\langle \sin\theta \rangle = \frac{1}{\pi} \int_0^\pi \sin\theta\, d\theta = \frac{1}{\pi} (-\cos\theta)\Big|_0^\pi = \frac{1}{\pi}(1-(-1)) = \frac{2}{\pi}$

$\langle \cos\theta \rangle = \frac{1}{\pi} \int \cos\theta\, d\theta = \frac{1}{\pi} \sin\theta\Big|_0^\pi = 0$

Note: $\langle \sin\theta \rangle = y/r$ $\quad \langle \cos\theta \rangle = x/r \Rightarrow \langle x \rangle = 0$ ok, $\langle y \rangle \neq 0$ ok

$\langle \cos^2\theta \rangle = \frac{1}{\pi} \int_0^\pi \cos^2\theta\, d\theta = \frac{1}{\pi} \frac{\pi}{2} = \frac{1}{2}$.

because $\cos^2\theta = \left(\frac{1}{2}(e^{i\theta} + e^{-i\theta})\right)^2 = \frac{1}{4}(e^{i2\theta} + e^{-i2\theta} + 2) = \frac{1}{2}(1 + \cos 2\theta)$

(5) $\psi(x,t) = A\, e^{-\lambda|x|}\, e^{-i\omega t}$

P1.5

(a) $1 = \int_{-\infty}^{\infty} dx\, |\psi(x,t)|^2 = A^2 \underbrace{|e^{-i\omega t}|^2}_{1} \int_{-\infty}^{\infty} e^{-2\lambda|x|}\, dx$

$= A^2 \cdot 2 \int_0^{\infty} dx\, e^{-2\lambda x} = \frac{2A^2}{2\lambda} = 1 \Rightarrow A = \sqrt{\lambda}$

(b) $\langle x \rangle = \lambda \int_{-\infty}^{\infty} x \left| e^{-\lambda|x|} e^{-i\omega t} \right|^2 dx =$

$= \lambda \int_{-\infty}^{\infty} dx\, x\, e^{-2\lambda|x|} = 0$ by oddness.

$\langle x^2 \rangle = \lambda \int_{-\infty}^{\infty} dx\, x^2 e^{-2\lambda|x|} =$

$= 2\lambda \int_0^{\infty} dx\, x^2 e^{-2\lambda x}$

To calculate $I_\lambda = \int_0^{\infty} dx\, x^2 e^{-2\lambda x}$ note that

$I_\lambda = \frac{1}{4} \frac{d^2}{d\lambda^2} \int_0^{\infty} dx\, e^{-2\lambda x} = \frac{1}{4} \int_0^{\infty} dx\, (-2x)^2 e^{-2\lambda x}$

Thus $I_\lambda = \frac{1}{4} \frac{d^2}{d\lambda^2} \frac{1}{2\lambda} = \frac{1}{8} \frac{d}{d\lambda} \frac{-1}{\lambda^2} = \frac{1}{4} \frac{1}{\lambda^3}$

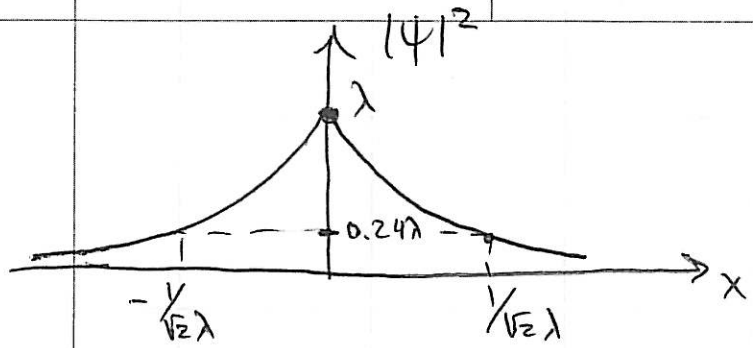
$\Rightarrow \langle x^2 \rangle = 2\lambda \cdot \frac{1}{4\lambda^3} = \frac{1}{2\lambda^2}$

(c) $\sigma^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{1}{2\lambda^2} - 0 = \frac{1}{2\lambda^2}$, $\sigma = \frac{1}{\sqrt{2}\lambda}$

$|\psi(x,t)|^2 = A^2 e^{-2\lambda|x|} = \lambda e^{-2\lambda|x|}$

$|\psi(x=\pm\sigma,t)|^2 = \lambda e^{-2\lambda/\sqrt{2}\lambda} = \lambda e^{-\sqrt{2}} \approx 0.24\lambda$

$|\psi(x=0,t)|^2 = \lambda$



Probability to be outside $(-\sigma, +\sigma)$ interval is

$$P = \int_{-\infty}^{-\sigma} |\psi|^2 dx + \int_{\sigma}^{\infty} |\psi|^2 dx =$$

$$= 2\int_{\sigma}^{\infty} dx\ \lambda e^{-2\lambda|x|} = -e^{-2\lambda x}\Big|_{\sigma}^{\infty} = e^{-2\lambda\sigma} =$$

$$= e^{-\sqrt{2}} \approx 0.24$$